Luis Fernando Pazzanese Pinheiro

Computer Engineering student at Insper — embedded systems, cybersecurity and low-level programming.

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EDUCATION

BSc in Computer Engineering · Insper — Feb 2024 — Dec 2028

Currently in the 6th semester; expected graduation Dec 2028.

- Coursework that maps to the target roles: Embedded Systems, Computer Architecture, Software Design, Data Science, Agile Development, Calculus I–III, Physics I–III.
- Top-ranked competitor in the internal CTFs of Insper SEC, the university's cybersecurity club — mostly web exploitation (SQLi, XSS, SSRF), with some crypto and misc.

EXPERIENCE

Technology Intern (Back Office) · Zagros Capital — Mar — Sep 2025

Independent real estate asset manager with R$2B+ under management, São Paulo. Internal tooling; code is private.

- Removed the manual consolidation of quarterly fund reports by writing Python scrapers that pull each report and merge its metrics into Excel sheets grouped by fund type (CRI-backed REITs, FIDCs and others).
- Kept fund values current with zero manual steps: a Streamlit dashboard whose long-polling job captures the previous day's NAV PDF every morning and updates the figures.
- Gave the team a single always-current view of holdings by building interactive per-asset charts and replicating them across the firm's entire fund portfolio.

PROJECTS

This site and a browser CTF · mywispy.com — May 2026 — present

- Built this site with Astro and a Three.js PCB-trace hero, statically generated to Cloudflare Pages; the EN/PT content and the resume PDF are rendered from one data file.
- Shipped /play, a four-level reverse-engineering challenge in a terminal UI — hidden files, base64, XOR and a final checker compiled from C to WebAssembly — with a Cloudflare Worker that validates flags, counts solvers in KV and rate-limits per IP.

Play the CTF ↗ · Source ↗

Embedded Systems — six open-source firmware projects — Feb — Jun 2026

· Insper · Raspberry Pi Pico 2 (RP2350)

Course projects on the RP2350 (dual Cortex-M33), all in C with the Pico SDK. The three below are the ones worth reading.

- **Air Mouse** — MPU6050 IMU sampled at 100 Hz over I2C, fused into orientation with the Madgwick-based Fusion AHRS and streamed as a 4-byte binary protocol to a Python host that moves the cursor. Four FreeRTOS tasks decoupled by three queues and one semaphore; a flick gesture clicks, with refractory-period debouncing. github.com/LuisPazzanese/pico-mpu-rtos
- **Touchscreen Stepper Controller** — Custom ILI9341 240×320 TFT driver plus 4-wire resistive touch read through the ADC (64-sample averaging, pressure gate), driving a unipolar stepper in half-step mode with an animated direction indicator. github.com/LuisPazzanese/pico-expert
- **Ultrasonic Ranger** — HC-SR04 ranging that never blocks: trigger via SDK alarms, echo timed with rising/falling-edge GPIO IRQs and time_us_64(), a 50 ms watchdog alarm flags lost echoes, and the period is reconfigurable over a serial CLI. github.com/LuisPazzanese/pico-timer

RTOS Multi-task System · ADC Pipeline · IRQ Buzzer

All repositories ↗

SKILLS

Languages	C · Python · x86 Assembly · JavaScript / TypeScript
Embedded	RP2350 (ARM Cortex-M33) · FreeRTOS · Pico SDK · I2C · SPI · UART · ADC · PWM · GPIO IRQs · GDB
Security	CTF web exploitation (SQLi, XSS, SSRF) · reverse engineering · Wireshark
Tools & web	Git · Linux · Python automation · Streamlit · Astro · WebAssembly · Cloudflare Workers
Spoken	Portuguese (native) · English (fluent — schooled in the UK and Canada)